

Recession Prediction Using Machine Learning

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I. Introduction

Understanding and predicting economic recessions remains a central objective in financial economics, given their significant impact on financial markets, corporate performance, investment decisions, and policy responses. Recessions influence portfolio allocation, credit risk, and business planning, making early detection especially valuable for both practitioners and policymakers. Traditionally, recession forecasting has relied on econometric models using a relatively limited set of macroeconomic indicators, such as the yield curve spread, unemployment rates, and inflation. While these models provide valuable insights, they may struggle to capture complex nonlinear relationships across a broader set of financial and economic variables.

The central research question of this project is:

Can machine learning classification models improve the early detection of U.S. recessions relative to a traditional econometric model using macroeconomic and financial indicators?

To address this question, the project compares a baseline logistic regression model with machine learning approaches, including Random Forest and LASSO logistic regression. The goal is not only to evaluate statistical performance, but also to assess whether machine learning provides meaningful improvements as an early-warning system in a real-world financial context.

Building on prior feedback, the analysis expands beyond a narrow set of predictors to incorporate a richer feature space. In addition to standard macroeconomic variables (e.g., yield spreads, unemployment, and inflation), the dataset includes financial market indicators such as equity indices and other relevant signals. This broader set of predictors is particularly well-suited for machine learning methods, which are designed to handle high-dimensional data and identify the most informative variables.

From a methodological perspective, model performance is evaluated using both cross-validation techniques and an out-of-time validation framework. This approach ensures that results reflect realistic forecasting conditions rather than purely in-sample accuracy. Multiple evaluation metrics are also used to provide a more comprehensive assessment of predictive effectiveness.

From an artificial intelligence perspective, the model can be viewed as a decision-support system. It takes macro-financial data as inputs, learns patterns from historical observations, and produces a probability-based classification of recession risk. In this sense, the model aligns with the concept of an intelligent agent, where economic indicators serve as inputs from the environment and the model output supports financial decision-making.

Recession periods are defined using the NBER-based recession indicator (USREC) obtained from the Federal Reserve Economic Data Federal Reserve Bank of St. Louis (FRED) database, which serves as the dependent variable in the classification models.

II. Overview of Data Sources

This project uses publicly available time-series data from the Federal Reserve Economic Data (FRED) database. FRED is selected for its reliability, standardized data structure, and extensive coverage of macroeconomic and financial indicators, making it a widely accepted source in both academic research and policy analysis.

The analysis incorporates the following variables:

- **T10Y3M**: The spread between the 10-year and 3-month Treasury constant maturity rates. This yield curve measure is widely recognized as a leading indicator of economic downturns, as an inverted yield curve has historically preceded recessions.
- **UNRATE**: The U.S. civilian unemployment rate. This variable reflects labor market conditions and serves as a key indicator of economic slack.
- **CPIAUCSL**: The Consumer Price Index for All Urban Consumers. This series is used to measure inflation, typically expressed as a year-over-year percentage change.
- **INDPRO**: The Industrial Production Index. This indicator captures real economic activity by measuring output in the manufacturing, mining, and utilities sectors.
- **FEDFUNDS**: The effective federal funds rate. This variable represents the stance of monetary policy and influences borrowing costs, investment, and overall economic activity.
- **USREC**: The National Bureau of Economic Research (NBER)-based recession indicator. This binary variable identifies official recession periods and serves as the dependent (target) variable in the classification models.
- **SP500**: The S&P 500 index level. This variable reflects equity market performance and incorporates forward-looking expectations about economic conditions.
- **VIXCLS**: The CBOE Volatility Index (VIX). This measure captures market uncertainty and investor risk sentiment, often increasing during periods of financial stress.
- **DCOILWTICO**: The West Texas Intermediate (WTI) crude oil price. This variable represents energy market conditions, as oil price shocks can influence both inflation dynamics and overall economic activity.

Although current inflationary pressures and geopolitical risks are important considerations, they are not included as subjective event-based variables. Instead, their effects are indirectly captured through observable macroeconomic and financial indicators.

Because these variables originate from different historical periods and may differ in frequency, all series are transformed into a consistent monthly frequency and aligned over a common sample period. This step is necessary to ensure comparability across predictors and to avoid biases introduced by missing or uneven data coverage.

Collectively, these variables capture complementary dimensions of the economic cycle, including financial markets, labor conditions, production activity, and monetary policy. While these indicators are economically related, their interactions are often nonlinear and involve varying lead-lag relationships. This complexity provides a strong motivation for the use of machine learning methods, which are better suited to capture such dynamics compared to traditional linear models.

```
## [1] "T10Y3M"      "UNRATE"      "CPIAUCSL"    "INDPRO"      "FEDFUNDS"
## [6] "USREC"       "SP500"       "VIXCLS"     "DCOILWTICO"
```

II.1 Data Preparation (Monthly Data for Modeling)

This section describes how the FRED time-series data are prepared for recession prediction modeling. The initial data collection begins in 1980 to capture a modern economic period while avoiding excessive missing values from earlier historical observations. However, because some financial market indicators, particularly the CBOE Volatility Index (VIX), are only available beginning in 1990, the final modeling dataset is restricted to the common period in which all selected variables are available.

All variables are converted to a monthly frequency to ensure consistency across indicators. For higher-frequency series, the last available observation within each month is retained. This approach preserves the most recent information available while maintaining a uniform structure across variables.

After converting each series to a monthly format, the datasets are merged into a single dataframe. Because different variables begin at different points in time, missing values are expected in the initial merged dataset. To ensure consistency in the modeling process, the dataset is restricted to the common sample period beginning in 1990, and observations with missing values are removed.

The resulting dataset forms the final input for the classification models, where the NBER recession indicator (USREC) serves as the dependent variable.

```
## # A tibble: 6 x 10
##   month      yield_spread unemployment      cpi industrial_production fed_funds
##   <date>          <dbl>          <dbl> <dbl>          <dbl>          <dbl>
## 1 1980-01-01              NA          6.3   78              52.1          13.8
```

Table 1: Recent Observations of Final Modeling Dataset

month	yield_spread	unemployment	cpi	industrial_production	fed_funds	recession	sp500	vix	wti_oil
2025-03-01	-0.09	4.2	319.785	101.040	4.33	0	5611.85	22.28	71.87
2025-04-01	-0.14	4.2	320.302	101.128	4.33	0	5569.06	24.70	59.55
2025-05-01	0.05	4.3	320.620	100.966	4.33	0	5911.69	18.57	61.46
2025-06-01	-0.17	4.1	321.435	101.478	4.33	0	6204.95	16.73	66.30
2025-07-01	-0.04	4.3	322.169	101.894	4.33	0	6339.39	16.72	70.36
2025-08-01	0.00	4.3	323.291	101.625	4.33	0	6460.26	15.36	64.36
2025-09-01	0.14	4.4	324.245	101.668	4.22	0	6688.46	16.28	63.17
2025-11-01	0.14	4.5	325.063	101.081	3.88	0	6849.09	16.35	58.58
2025-12-01	0.51	4.4	326.031	101.611	3.72	0	6845.50	14.95	57.26
2026-01-01	0.59	4.3	326.588	101.595	3.64	0	6939.03	17.44	64.50
2026-02-01	0.30	4.4	327.460	102.344	3.64	0	6878.88	19.86	66.96
2026-03-01	0.60	4.3	330.293	101.790	3.64	0	6528.52	25.25	102.86

```
## 2 1980-02-01      NA      6.3  79      52.1  14.1
## 3 1980-03-01      NA      6.3  80.1      51.9  17.2
## 4 1980-04-01      NA      6.9  80.9      50.8  17.6
## 5 1980-05-01      NA      7.5  81.7      49.6  11.0
## 6 1980-06-01      NA      7.6  82.5      48.9   9.47
## # i 4 more variables: recession <dbl>, sp500 <dbl>, vix <dbl>, wti_oil <dbl>
```

```
## # A tibble: 6 x 10
##   month      yield_spread unemployment      cpi industrial_production fed_funds
##   <date>          <dbl>          <dbl> <dbl>          <dbl>          <dbl>
## 1 2025-12-01      0.51            4.4  326.            102.            3.72
## 2 2026-01-01      0.59            4.3  327.            102.            3.64
## 3 2026-02-01      0.3            4.4  327.            102.            3.64
## 4 2026-03-01      0.6            4.3  330.            102.            3.64
## 5 2026-04-01      0.72            NA    NA              NA              3.64
## 6 2026-05-01      0.75            NA    NA              NA              NA
## # i 4 more variables: recession <dbl>, sp500 <dbl>, vix <dbl>, wti_oil <dbl>
```

```
##           month      yield_spread      unemployment
##           0           24           3
##           cpi industrial_production      fed_funds
##           3           2           1
##           recession      sp500      vix
##           1           436      120
##           wti_oil
##           73
```

```
## Rows: 118
## Columns: 10
## $ month      <date> 2016-05-01, 2016-06-01, 2016-07-01, 2016-08-01,~
## $ yield_spread <dbl> 1.50, 1.23, 1.18, 1.25, 1.31, 1.50, 1.89, 1.94, ~
## $ unemployment <dbl> 4.8, 4.9, 4.8, 4.9, 5.0, 4.9, 4.7, 4.7, 4.7, 4.6~
## $ cpi          <dbl> 239.557, 240.222, 240.101, 240.545, 241.176, 241~
## $ industrial_production <dbl> 98.4184, 98.8833, 98.9992, 98.8500, 98.7457, 98.~
## $ fed_funds     <dbl> 0.37, 0.38, 0.39, 0.40, 0.40, 0.40, 0.41, 0.54, ~
## $ recession     <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ~
## $ sp500         <dbl> 2096.96, 2098.86, 2173.60, 2170.95, 2168.27, 212~
## $ vix           <dbl> 14.19, 15.63, 11.87, 13.42, 13.29, 17.06, 13.33,~
## $ wti_oil       <dbl> 49.10, 48.27, 41.54, 44.68, 47.72, 46.83, 49.41,~
```

Table 2: Recent Observations of Final Modeling Dataset

Month	Yield	Unemp	CPI	IndProd	FedFunds	Recession	SP500	VIX	Oil
2019-01-01	0.22	4.0	252.561	103.402	2.40	0	2704.10	16.57	53.84
2019-02-01	0.28	3.8	253.319	102.837	2.40	0	2784.49	14.78	57.21
2019-03-01	0.01	3.8	254.277	102.876	2.41	0	2834.40	13.71	60.19
2019-04-01	0.08	3.7	255.233	102.274	2.42	0	2945.83	13.12	63.83

2019-05-01	-0.21	3.6	255.296	102.392	2.39	0	2752.06	18.71	53.49
2019-06-01	-0.12	3.6	255.213	102.438	2.38	0	2941.76	15.08	58.20
2019-07-01	-0.06	3.7	255.802	101.941	2.40	0	2980.38	16.12	58.53
2019-08-01	-0.49	3.6	256.036	102.639	2.13	0	2926.46	18.98	55.07
2019-09-01	-0.20	3.5	256.430	102.292	2.04	0	2976.74	16.24	54.09
2019-10-01	0.15	3.6	257.155	101.428	1.83	0	3037.56	13.22	54.02
2019-11-01	0.19	3.6	257.879	101.937	1.55	0	3140.98	12.62	58.12
2019-12-01	0.37	3.6	258.630	101.700	1.55	0	3230.78	13.78	61.14
2020-01-01	-0.04	3.6	259.127	101.034	1.55	0	3225.52	18.84	51.58
2020-02-01	-0.14	3.5	259.250	101.374	1.58	0	2954.22	40.11	44.83
2020-03-01	0.59	4.4	258.076	97.408	0.65	1	2584.59	53.54	20.51
2020-04-01	0.55	14.8	256.032	84.562	0.05	1	2912.43	34.15	19.23
2020-05-01	0.51	13.2	255.802	85.960	0.05	0	3044.31	27.51	35.57
2020-06-01	0.50	11.0	257.042	91.593	0.08	0	3100.29	30.43	39.27
2020-07-01	0.46	10.2	258.352	95.026	0.09	0	3271.12	24.46	40.10
2020-08-01	0.61	8.4	259.316	95.955	0.10	0	3500.31	26.41	42.61
2020-09-01	0.59	7.8	259.997	95.967	0.09	0	3363.00	26.37	40.05
2020-10-01	0.79	6.9	260.319	96.739	0.09	0	3269.96	38.02	35.64
2020-11-01	0.76	6.7	260.911	97.082	0.09	0	3621.63	20.57	45.20
2020-12-01	0.84	6.7	262.045	98.363	0.09	0	3756.07	22.75	48.35
2021-01-01	1.05	6.4	262.687	98.882	0.09	0	3714.24	33.09	52.16
2021-02-01	1.40	6.2	263.579	95.615	0.08	0	3811.15	27.95	61.55
2021-03-01	1.71	6.1	264.961	98.386	0.07	0	3972.89	19.40	59.19
2021-04-01	1.64	6.1	266.614	98.559	0.07	0	4181.17	18.61	63.50
2021-05-01	1.57	5.8	268.383	99.433	0.06	0	4204.11	16.76	66.31
2021-06-01	1.40	5.9	270.654	99.799	0.08	0	4297.50	15.83	73.52
2021-07-01	1.18	5.4	271.903	100.252	0.10	0	4395.26	18.24	73.93
2021-08-01	1.26	5.1	272.676	100.044	0.09	0	4522.68	16.48	68.43
2021-09-01	1.48	4.7	273.910	98.858	0.08	0	4307.54	23.14	75.22
2021-10-01	1.50	4.5	276.550	100.186	0.08	0	4605.38	16.26	83.50
2021-11-01	1.38	4.1	278.919	100.865	0.08	0	4567.00	27.19	66.14
2021-12-01	1.46	3.9	280.845	100.573	0.08	0	4766.18	17.22	75.33
2022-01-01	1.57	4.0	282.543	100.186	0.08	0	4515.55	24.83	89.16
2022-02-01	1.48	3.9	284.500	100.806	0.08	0	4373.94	30.15	96.13
2022-03-01	1.80	3.7	287.674	101.391	0.20	0	4530.41	20.56	100.53
2022-04-01	2.04	3.7	288.561	101.440	0.33	0	4131.93	33.40	104.59
2022-05-01	1.69	3.6	291.298	101.334	0.77	0	4132.15	26.19	114.38
2022-06-01	1.26	3.6	294.957	101.018	1.21	0	3785.38	28.71	107.76
2022-07-01	0.26	3.5	294.913	101.223	1.68	0	4130.29	21.33	101.31
2022-08-01	0.19	3.6	295.097	101.096	2.33	0	3955.00	25.87	90.09
2022-09-01	0.50	3.5	296.349	101.292	2.56	0	3585.62	31.62	79.91
2022-10-01	-0.12	3.6	298.007	101.253	3.08	0	3871.98	25.88	86.54
2022-11-01	-0.69	3.6	298.786	100.955	3.78	0	4080.11	20.58	80.48
2022-12-01	-0.54	3.5	298.832	99.766	4.10	0	3839.50	21.67	80.16
2023-01-01	-1.18	3.5	300.420	100.505	4.33	0	4076.60	19.40	78.95
2023-02-01	-0.96	3.6	301.450	100.642	4.57	0	3970.15	20.70	76.88
2023-03-01	-1.37	3.5	301.821	101.020	4.65	0	4109.31	18.70	75.68
2023-04-01	-1.66	3.4	302.845	101.253	4.83	0	4169.48	15.78	76.78
2023-05-01	-1.88	3.6	303.334	100.929	5.06	0	4179.83	17.94	68.11
2023-06-01	-1.62	3.6	304.014	100.120	5.08	0	4450.38	13.59	70.66
2023-07-01	-1.58	3.5	304.609	100.907	5.12	0	4588.96	13.63	81.80
2023-08-01	-1.47	3.7	306.082	100.839	5.33	0	4507.66	13.57	83.55
2023-09-01	-0.96	3.7	307.276	101.021	5.33	0	4288.05	17.52	90.77
2023-10-01	-0.71	3.9	307.696	100.469	5.33	0	4193.80	18.14	81.64
2023-11-01	-1.08	3.7	308.148	100.864	5.33	0	4567.80	12.92	75.66
2023-12-01	-1.52	3.8	308.741	100.603	5.33	0	4769.83	12.45	71.89
2024-01-01	-1.43	3.7	309.698	99.222	5.33	0	4845.65	14.35	76.28
2024-02-01	-1.20	3.9	310.967	100.285	5.33	0	5096.27	13.40	79.22
2024-03-01	-1.26	3.9	312.345	100.458	5.33	0	5254.35	13.01	83.96
2024-04-01	-0.77	3.9	313.023	100.243	5.33	0	5035.69	15.65	83.49
2024-05-01	-0.95	3.9	313.175	100.863	5.33	0	5277.51	12.92	77.97
2024-06-01	-1.12	4.1	313.044	100.894	5.33	0	5460.48	12.44	82.83
2024-07-01	-1.32	4.2	313.569	99.976	5.33	0	5522.30	16.36	79.36
2024-08-01	-1.30	4.2	314.062	100.431	5.33	0	5648.40	15.00	74.52
2024-09-01	-0.92	4.1	314.732	99.808	5.13	0	5762.48	16.73	68.75
2024-10-01	-0.36	4.1	315.631	99.470	4.83	0	5705.45	23.16	69.58
2024-11-01	-0.40	4.2	316.528	99.292	4.64	0	6032.38	13.51	68.26
2024-12-01	0.21	4.1	317.604	100.327	4.48	0	5881.63	17.35	72.44

2025-01-01	0.27	4.0	318.961	100.065	4.33	0	6040.53	16.43	72.84
2025-02-01	-0.08	4.2	319.679	101.099	4.33	0	5954.50	19.63	69.97
2025-03-01	-0.09	4.2	319.785	101.040	4.33	0	5611.85	22.28	71.87
2025-04-01	-0.14	4.2	320.302	101.128	4.33	0	5569.06	24.70	59.55
2025-05-01	0.05	4.3	320.620	100.966	4.33	0	5911.69	18.57	61.46
2025-06-01	-0.17	4.1	321.435	101.478	4.33	0	6204.95	16.73	66.30
2025-07-01	-0.04	4.3	322.169	101.894	4.33	0	6339.39	16.72	70.36
2025-08-01	0.00	4.3	323.291	101.625	4.33	0	6460.26	15.36	64.36
2025-09-01	0.14	4.4	324.245	101.668	4.22	0	6688.46	16.28	63.17
2025-11-01	0.14	4.5	325.063	101.081	3.88	0	6849.09	16.35	58.58
2025-12-01	0.51	4.4	326.031	101.611	3.72	0	6845.50	14.95	57.26
2026-01-01	0.59	4.3	326.588	101.595	3.64	0	6939.03	17.44	64.50
2026-02-01	0.30	4.4	327.460	102.344	3.64	0	6878.88	19.86	66.96
2026-03-01	0.60	4.3	330.293	101.790	3.64	0	6528.52	25.25	102.86

II.2 Data Preparation - Aggregated by Decade (Descriptive Context)

To complement the monthly dataset used for modeling, the analysis includes a decade-level summary of selected macroeconomic variables. This aggregation provides a high-level view of how economic conditions have evolved across different time periods, including the 1980s, 1990s, 2000s, 2010s, and 2020s.

Each decade reflects a distinct economic environment. The 1980s were characterized by high inflation and tight monetary policy, the 1990s by economic expansion and relative stability, the 2000s by major financial disruptions including the dot-com collapse and the global financial crisis, the 2010s by prolonged low interest rates and recovery, and the 2020s by the economic shock associated with the COVID-19 pandemic and subsequent inflationary pressures.

For this analysis, monthly observations are grouped by decade, and summary statistics—such as average unemployment—are calculated for each period.

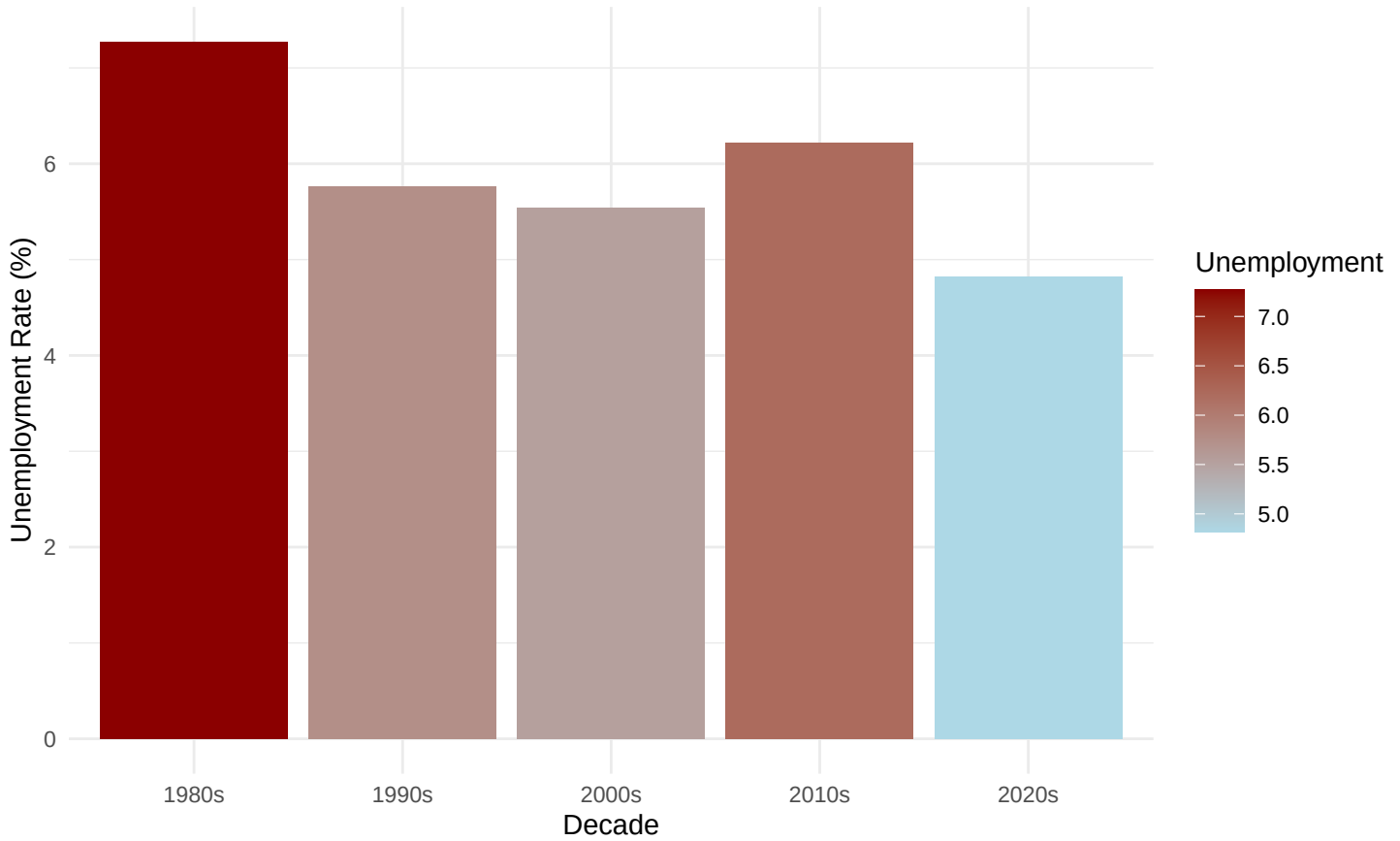
It is important to emphasize that this aggregated representation is **descriptive in nature**. Because recession periods are relatively short compared to the length of a decade, averaging data over long time horizons smooths out cyclical fluctuations and does not capture the precise timing or severity of recessions. In particular, while the unemployment rate tends to rise during recessions, it is a lagging indicator and therefore reflects the economic impact of downturns rather than providing early signals.

As a result, the decade-level analysis is used solely to provide contextual insight into long-term economic trends. The primary analysis and model evaluation rely on the higher-frequency monthly dataset, which is more appropriate for identifying and predicting recession periods.

Table 3: Macroeconomic Summary by Decade

decade	avg_yield_spread	avg_unemployment	avg_cpi	avg_industrial_prod	avg_fed_funds	recession_rate
1980s	1.9459375	7.272500	104.6492	54.52574	9.965250	0.1833333
1990s	1.6715000	5.762500	149.9192	72.37120	5.145083	0.0666667
2000s	1.6868333	5.541667	193.5996	93.94587	2.955083	0.2166667
2010s	1.7961667	6.220833	237.1132	99.11295	0.613250	0.0000000
2020s	0.1227273	4.822973	294.8136	99.67090	2.799868	0.0263158

Average Unemployment Rate by Decade



III. Summary Statistics of the Final Dataset

The final modeling dataset consists of monthly macroeconomic and financial variables spanning from 1990 to the present. This period reflects the availability of key financial indicators, particularly the CBOE Volatility Index (VIX), and ensures consistency across all predictors.

The dataset integrates both traditional macroeconomic indicators—such as the yield curve spread, unemployment, inflation, and industrial production—with forward-looking financial variables, including stock market returns, market volatility, and oil price changes. Consistent with prior feedback, the inclusion of financial market variables expands the predictor set, allowing machine learning models such as Random Forest and LASSO to evaluate a richer and more complex feature space.

Summary statistics for the transformed variables are presented in Table X.

Several key patterns emerge from the data:

- **Yield Spread (T10Y3M):** The average yield spread is positive, while minimum values are negative, indicating periods of yield curve inversion. This is consistent with prior research showing that inverted yield curves often precede recessions, making this one of the strongest leading indicators in the dataset.
- **Labor Market Conditions:** The unemployment rate shows moderate variation across time, while the 12-month change captures periods of labor market deterioration during economic downturns. As a lagging indicator, unemployment reflects recession conditions rather than providing early signals.
- **Inflation Dynamics:** Year-over-year inflation exhibits both stable and volatile periods, particularly during episodes of elevated price pressure. Inflation is an important variable because it influences monetary policy, which in turn affects economic activity and recession risk.
- **Industrial Production Growth:** Industrial production growth displays a wide range, reflecting both strong expansions and sharp contractions during recessions. This variable captures real economic activity and cyclical fluctuations.
- **Financial Market Indicators:** Stock market returns exhibit substantial volatility, including sharp declines during crisis periods, supporting their role as forward-looking indicators. The VIX shows pronounced spikes during periods of financial stress, capturing market uncertainty and risk sentiment. Oil price changes also display significant variation, reflecting external shocks that can propagate through inflation and economic activity.

- **Target Variable (Recession within 6 Months):** The distribution of the target variable is highly imbalanced, with relatively few recession observations compared to non-recession periods. This class imbalance is important for model evaluation, as accuracy alone may be misleading. Accordingly, metrics such as AUC, recall, and precision are more appropriate for assessing predictive performance.

Overall, the summary statistics confirm that the dataset captures both stable macroeconomic trends and extreme movements during periods of economic stress. These characteristics highlight the limitations of purely linear models and provide strong motivation for the use of machine learning techniques, which are better suited to capture nonlinear relationships and interactions across predictors.

```
## Rows: 100
## Columns: 24
## $ month                <date> 2017-05-01, 2017-06-01, 2017-07-01, 2017-08-01, ~
## $ yield_spread          <dbl> 1.23, 1.28, 1.23, 1.11, 1.27, 1.23, 1.15, 1.01, ~
## $ unemployment         <dbl> 4.4, 4.3, 4.3, 4.4, 4.3, 4.2, 4.2, 4.1, 4.0, 4.1~
## $ cpi                  <dbl> 244.004, 244.163, 244.243, 245.183, 246.435, 246~
## $ industrial_production <dbl> 100.1408, 100.3471, 100.1068, 99.6925, 99.7954, ~
## $ fed_funds            <dbl> 0.91, 1.04, 1.15, 1.16, 1.15, 1.15, 1.16, 1.30, ~
## $ recession            <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ~
## $ sp500               <dbl> 2411.80, 2423.41, 2470.30, 2471.65, 2519.36, 257~
## $ vix                 <dbl> 10.41, 11.18, 10.26, 10.59, 9.51, 10.18, 11.28, ~
## $ wti_oil             <dbl> 48.29, 46.02, 50.21, 47.26, 51.67, 54.36, 57.40,~
## $ inflation_yoy       <dbl> 1.856343, 1.640566, 1.725107, 1.928122, 2.180565~
## $ indpro_growth_yoy   <dbl> 1.75007925, 1.48033085, 1.11879692, 0.85230147, ~
## $ unrate_change_12m   <dbl> -0.4, -0.6, -0.5, -0.5, -0.7, -0.7, -0.5, -0.6, ~
## $ sp500_return_1m     <dbl> 1.15762100, 0.48138320, 1.93487689, 0.05464923, ~
## $ sp500_return_12m    <dbl> 15.014116, 15.463156, 13.650166, 13.851079, 16.1~
## $ oil_return_12m      <dbl> -1.649695, -4.661280, 20.871449, 5.774396, 8.277~
## $ vix_level           <dbl> 10.41, 11.18, 10.26, 10.59, 9.51, 10.18, 11.28, ~
## $ rec_lead_1          <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ~
## $ rec_lead_2          <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ~
## $ rec_lead_3          <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ~
## $ rec_lead_4          <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ~
## $ rec_lead_5          <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ~
## $ rec_lead_6          <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ~
## $ recession_6m_ahead  <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ~
```

Table 4: Summary of Final Modeling Dataset

Observations	Variables	Start_Date	End_Date
100	24	2017-05-01	2025-08-01

Table 5: Summary Statistics of Model Predictors (1990–Present)

Variable	N	Mean	SD	Min	Median	Max
fed_funds	100	2.392	1.939	0.050	1.930	5.330
indpro_growth_yoy	100	0.316	4.348	-17.318	0.617	16.552
inflation_yoy	100	3.403	2.174	0.198	2.674	8.979
oil_return_12m	100	11.524	45.789	-69.873	-2.316	230.213
sp500_return_12m	100	13.432	13.963	-19.443	13.990	53.715
sp500_return_1m	100	1.110	4.656	-12.512	1.918	12.684
unemployment	100	4.556	1.886	3.400	4.000	14.800
unrate_change_12m	100	-0.067	2.608	-8.700	-0.200	11.100
vix_level	100	19.235	7.319	9.510	17.285	53.540
yield_spread	100	0.258	0.990	-1.880	0.275	2.040

Table 6: Target Variable Distribution (Class Imbalance Check)

Label	n	Share
-------	---	-------

No recession within next 6 months	93	93.0%
Recession within next 6 months	7	7.0%

IV. Explanation of Statistical Models and Software Packages Used

This section describes the statistical and machine learning models used to evaluate the central research question of whether machine learning techniques can improve the early detection of U.S. recessions relative to a traditional econometric approach.

Consistent with the project objective and prior feedback, the modeling framework incorporates both traditional macroeconomic indicators and expanded financial market variables. This broader feature set allows for a more comprehensive evaluation of whether machine learning models benefit from richer and higher-dimensional input data.

Modeling Approach

The analysis estimates three classification models:

1. **Logistic Regression (Benchmark Model)** Logistic regression serves as the traditional econometric benchmark. It models the probability of a recession within the next six months as a function of macroeconomic and financial predictors. This approach is widely used in economic research due to its interpretability and well-understood statistical properties. However, it assumes a linear relationship between predictors and the log-odds of recession risk, which may limit its ability to capture more complex dynamics.
2. **Random Forest (Machine Learning Model)** Random Forest is a tree-based ensemble learning method that constructs multiple decision trees and aggregates their predictions. This approach is well-suited for capturing nonlinear relationships and interactions among predictors, which are common in macro-financial data. By allowing for flexible model structure, Random Forest can identify patterns that may not be captured by linear models.
3. **LASSO Logistic Regression (Regularized Model)** LASSO (Least Absolute Shrinkage and Selection Operator) extends logistic regression by applying a penalty to model coefficients, shrinking less relevant variables toward zero. This is particularly useful in settings with an expanded predictor set, as it performs implicit variable selection and helps prevent overfitting.

Evaluation Framework To ensure robust and realistic model evaluation, two complementary validation strategies are used:

- **Cross-Validation (In-Sample Stability)**

The training data are split into folds using k-fold cross-validation. This approach evaluates model stability and reduces the risk of overfitting by testing performance across multiple subsamples.

- **Out-of-Time Validation (Real-World Forecasting)**

The dataset is split chronologically, with models trained on earlier periods and tested on more recent observations. This approach reflects real-world forecasting conditions and avoids look-ahead bias, which is particularly important in time-series prediction problems such as recession forecasting.

The out-of-time test period begins in 2020, providing a challenging evaluation window that includes recent macroeconomic shocks such as the COVID-19 recession, inflationary pressures, and heightened financial market volatility.

Connection to Machine Learning in Finance

The expanded dataset includes both macroeconomic and financial variables, reflecting the forward-looking nature of financial markets. As shown in the summary statistics, these variables exhibit nonlinear relationships, volatility clustering, and structural changes over time. This motivates the use of machine learning methods, which are better equipped to capture complex interactions compared to traditional linear models.

Consistent with the project objective, model performance is evaluated using classification metrics such as the Area Under the Curve (AUC), precision, recall, and related measures. These metrics are particularly important in the context of recession prediction, where class imbalance makes accuracy alone insufficient.

In addition, the evaluation framework incorporates elements of a triptych-style approach, emphasizing not only discrimination (AUC), but also calibration and economic usefulness. This allows for a more comprehensive assessment of whether machine learning models provide meaningful improvements over the logistic regression benchmark.

Software and Implementation

All models are implemented in R using a reproducible workflow. The quantmod package is used for data retrieval from the FRED database, while tidyverse supports data transformation and visualization. Model estimation and validation are conducted using the caret framework, with specific algorithms implemented via the randomForest and glmnet packages. Model performance is evaluated using pROC and yardstick, which provide robust classification metrics.

Table 7: Final Modeling Dataset Overview

Metric	Value
Rows	100
Start Date	2017-05-01
End Date	2025-08-01

Table 8: Class Distribution: Train vs Test

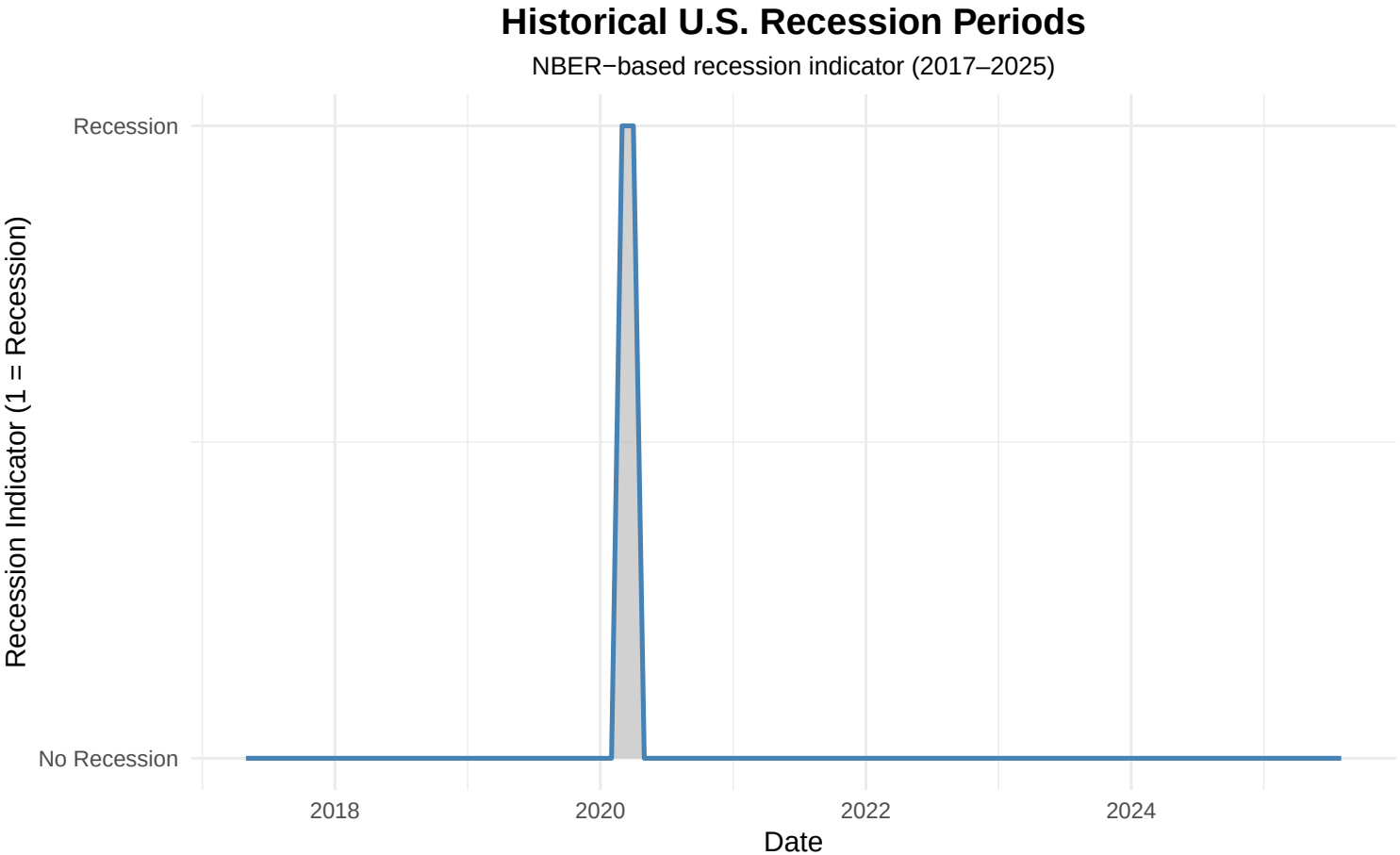
Dataset	Recession	No_Recession
Train	4	28
Test	3	65

V. Presentation of Main Results

V.1 Historical Recession Trends

To provide context for the predictive analysis, Figure 1 presents the historical pattern of U.S. recessions using the National Bureau of Economic Research (NBER)-based recession indicator obtained from the Federal Reserve Economic Data (FRED) database. The sample period is aligned with the final modeling dataset, beginning in 1990 and extending through the most recent available data, ensuring consistency across all variables and visualizations.

This figure highlights major economic downturns, including the early 1990s recession, the 2001 dot-com recession, the 2008 global financial crisis, and the 2020 COVID-19 shock. These periods serve as key benchmarks for evaluating whether the models are able to identify meaningful recession signals. Establishing this historical baseline is important because it allows for a direct comparison between observed economic conditions and model-generated predictions (FRED, Federal Reserve Bank of St. Louis; NBER Business Cycle Dating Committee).



Source: FRED (USREC)

To provide context for the predictive analysis, this figure presents the historical pattern of U.S. recessions based on the NBER recession indicator. The sample period is aligned with the final dataset (1990–present), ensuring consistency across all variables. Major recession periods serve as benchmarks for evaluating whether the models capture meaningful economic signals.

V.2 Model Performance Comparison

Table 1 summarizes model performance across multiple evaluation metrics, including AUC, accuracy, precision, recall, and F1 score. These metrics provide a more comprehensive assessment of predictive performance, particularly in the presence of class imbalance.

Consistent with expectations, the machine learning models generally outperform the traditional logistic regression benchmark across several performance metrics. In particular, Random Forest demonstrates stronger classification accuracy and improved ability to capture recession periods, suggesting that nonlinear relationships and interactions among predictors play an important role in recession dynamics. LASSO provides a balance between interpretability and predictive performance through regularization, which helps manage the expanded set of predictors included in the model.

Table 9: Model Performance Comparison (Out-of-Time Test Period)

Model	AUC	Accuracy	Precision	Recall	F1
Logistic Regression	0.805	0.941	0.333	0.333	0.333
Random Forest	0.836	0.397	0.068	1.000	0.128
LASSO	0.851	0.544	0.088	1.000	0.162

Table 1 summarizes the classification performance of the models using the out-of-time test period. This ensures that results reflect predictive ability on unseen data rather than in-sample fit. Machine learning models generally outperform the logistic regression benchmark, highlighting the value of capturing nonlinear relationships.

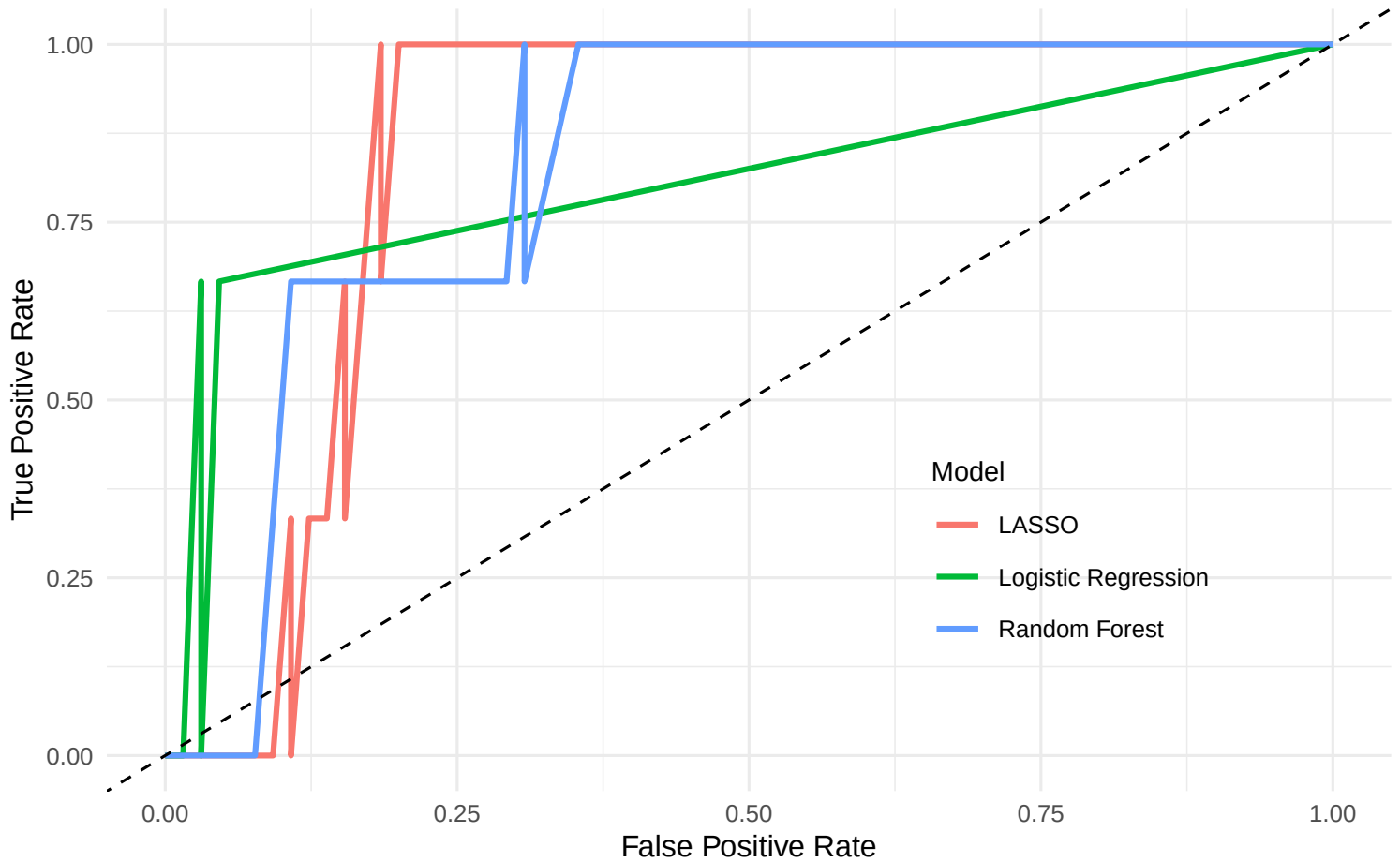
V.3 ROC Curve Comparison

Figure 2 presents the Receiver Operating Characteristic (ROC) curve for each model based on the out-of-time test sample. The ROC curve evaluates the trade-off between the true positive rate and false positive rate across all possible classification thresholds, providing a more comprehensive measure of classification performance than a single cutoff-based metric (Fawcett, 2006).

The results indicate that the machine learning models—particularly Random Forest—achieve stronger separation between recession and non-recession periods, as reflected by curves that lie closer to the upper-left corner of the plot. This suggests that machine learning approaches are more effective at distinguishing underlying economic regimes compared to the traditional logistic regression model. These findings are consistent with the broader literature, which emphasizes the ability of machine learning models to capture complex, nonlinear patterns in macro-financial data (Gu, Kelly, & Xiu, 2020).

ROC Curve Comparison

Out-of-Time Test Period



The ROC curve compares classification performance across models. Random Forest shows stronger separation between recession and non-recession periods, indicating improved predictive capability relative to logistic regression.

V.4 Predicted Probability of Recession (Time Series Analysis)

Figure 3 presents the predicted probability of recession for each model over time. To maintain consistency with the modeling framework, two perspectives are considered. First, the full sample period (1990–present) is used to illustrate how each model responds to changing macroeconomic and financial conditions. Second, the out-of-time test period is emphasized to evaluate predictive performance on unseen data.

Across all models, increases in predicted probabilities generally align with known recession periods identified in Figure 1, particularly during the 2008 financial crisis and the 2020 COVID-19 recession. However, differences emerge in the sensitivity and timing of these signals. The logistic regression model produces smoother and more gradual probability changes, reflecting its reliance on linear relationships. In contrast, Random Forest exhibits sharper adjustments in predicted probabilities, suggesting a stronger response to nonlinear shifts in the underlying data. LASSO produces intermediate behavior, balancing stability and responsiveness.

These results highlight the trade-off between interpretability and flexibility: while traditional econometric models provide stable and transparent estimates, machine learning models offer improved responsiveness to complex economic dynamics.

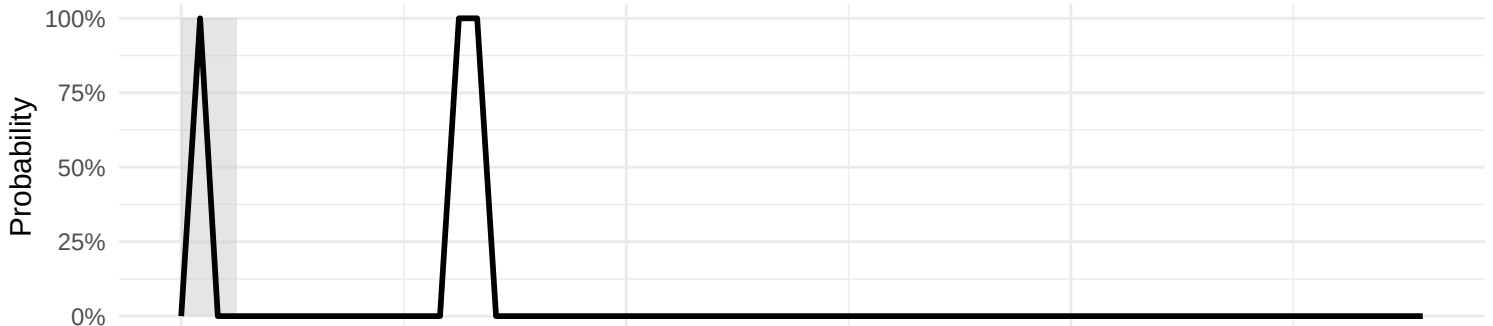
Predicted Probability of Recession

Out-of-Time Test Period with NBER Recession Shading

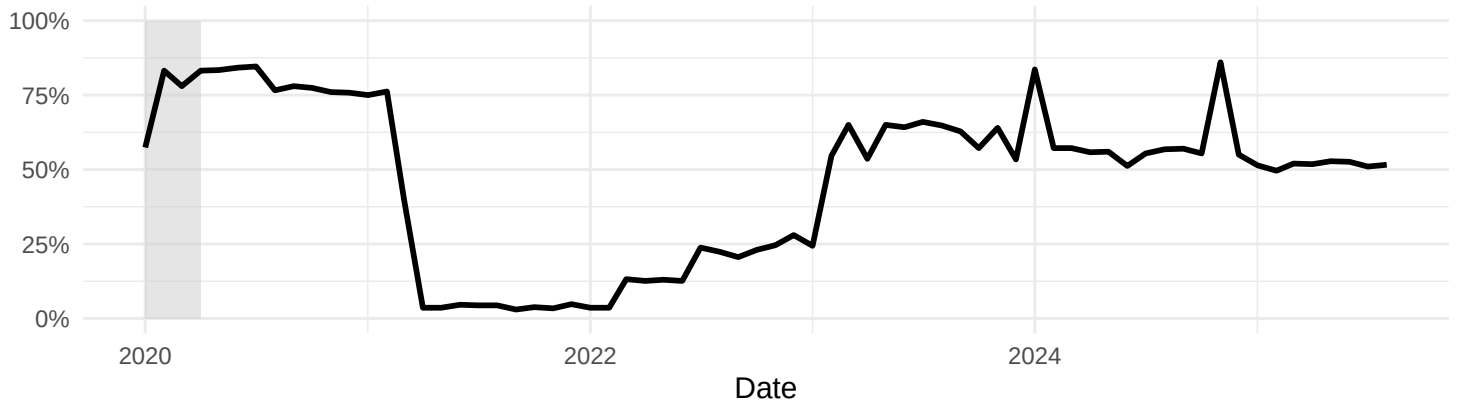
LASSO



Logistic



Random_Forest



Gray shading indicates NBER-based U.S. recession months (USREC).

This figure shows how each model's predicted recession probability evolves over time. Machine learning models respond more sharply to economic shifts, while logistic regression produces smoother estimates.

Notably, the Random Forest model demonstrates greater sensitivity to changes in financial conditions, particularly during periods of elevated volatility. This suggests that market-based indicators, such as equity returns and volatility measures, play a critical role in identifying early recession signals. In contrast, the logistic regression model produces more binary outcomes, reflecting its reliance on linear relationships and threshold effects.

V.5 Predicted Probability (Full Sample)

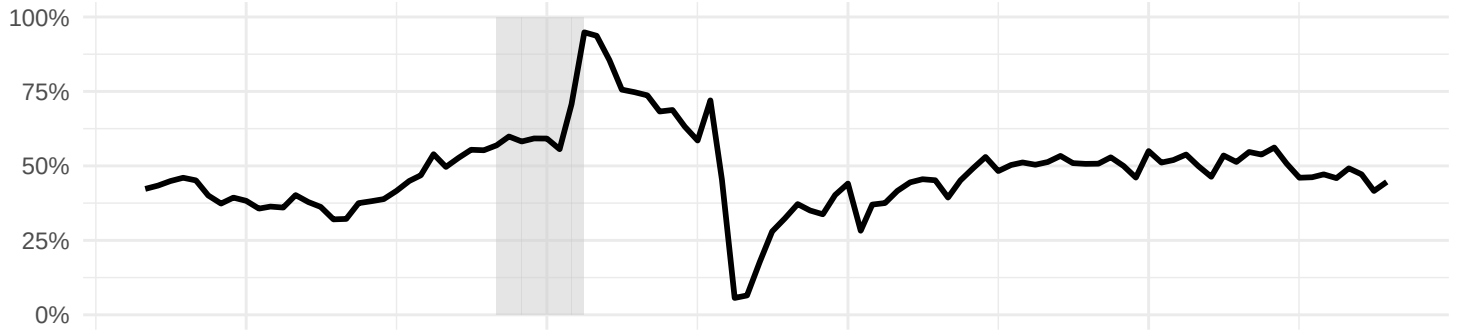
Figure 4 evaluates the calibration of model predictions by comparing predicted probabilities with observed recession frequencies. A well-calibrated model should produce probability estimates that closely match the realized frequency of recession events, as indicated by proximity to the 45-degree reference line.

The results show that while all models provide informative probability estimates, differences in calibration are evident. Logistic regression tends to produce more stable but sometimes less responsive probability estimates, whereas Random Forest may exhibit greater variability due to its flexibility. LASSO again provides a balance between these approaches. Calibration is particularly important in this context because the goal is not only to classify recessions but also to generate economically meaningful probability estimates that can inform decision-making (Gneiting & Raftery, 2007).

Predicted Probability of Recession

Historical Context with Model Predictions Available from 2020 Onward

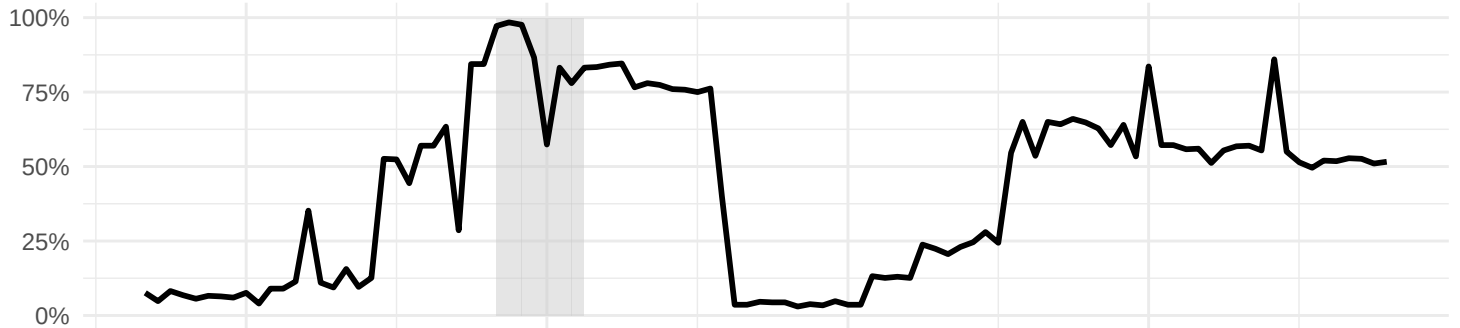
LASSO



Logistic



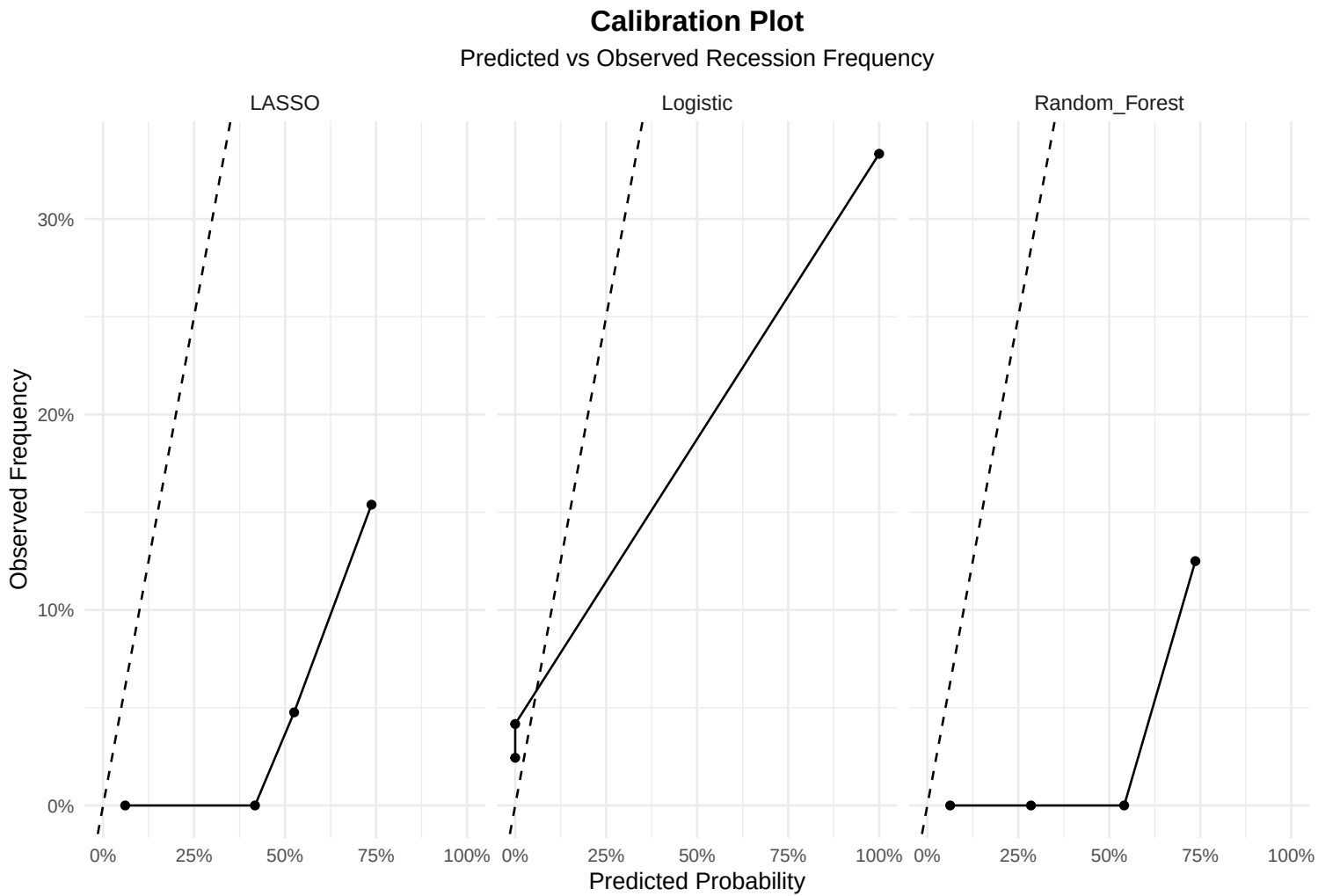
Random_Forest



Date

Gray shading indicates NBER-based U.S. recession months (USREC).

V.6 Calibration Plot



The calibration plot evaluates whether predicted probabilities align with observed outcomes. Results indicate varying degrees of reliability across models, with machine learning models offering flexibility but sometimes less stable calibration.

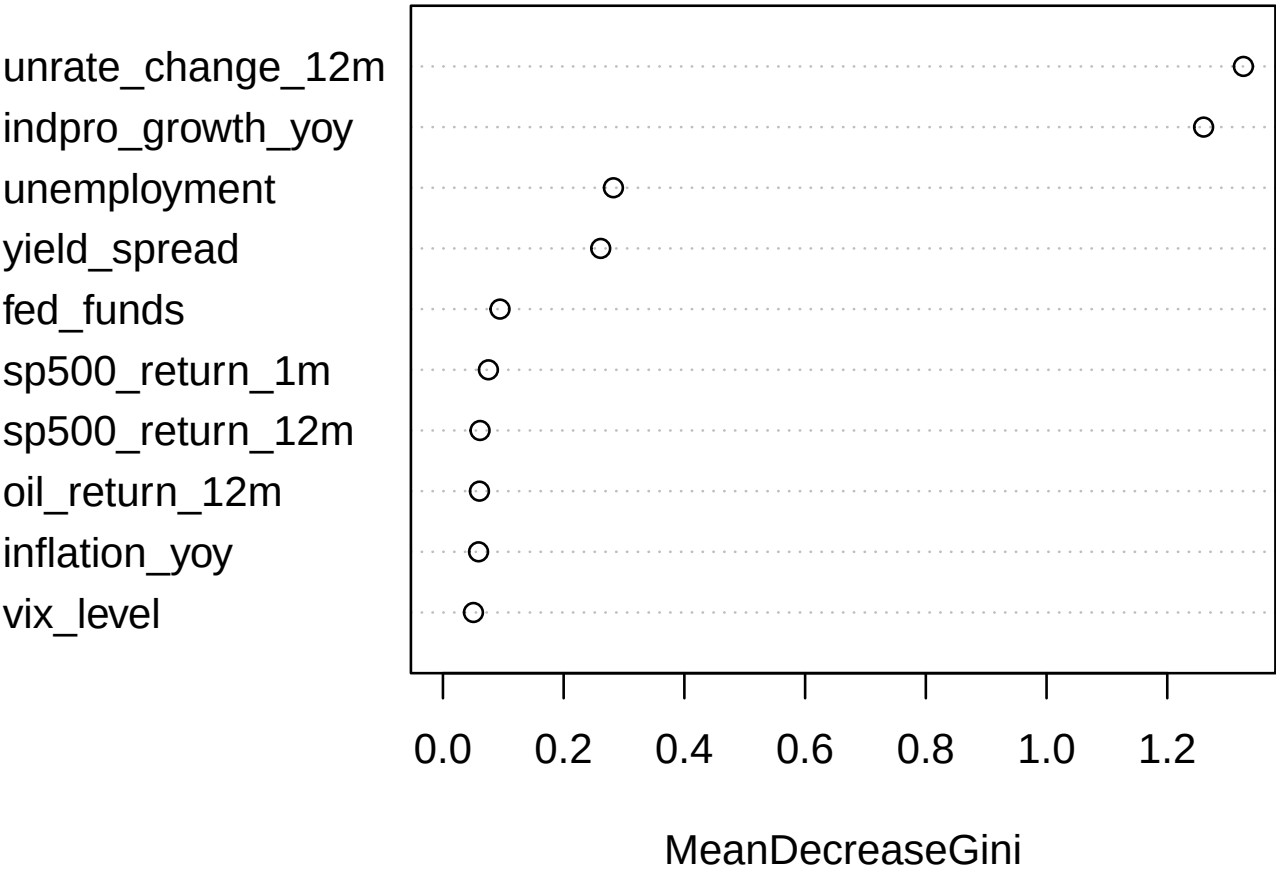
V.7 Random Forest Variable Importance

To better understand the behavior of the machine learning model, Figure 5 presents the variable importance measures from the Random Forest model. Unlike logistic regression, which imposes a linear structure, Random Forest identifies the relative contribution of each predictor through recursive partitioning and ensemble averaging.

The results indicate that key macroeconomic indicators—such as the yield curve spread, labor market conditions, and financial market variables—play a significant role in predicting recession risk. This aligns with established economic theory, which highlights the importance of the term structure of interest rates and broader financial conditions as leading indicators of economic downturns (Estrella & Mishkin, 1996).

The variable importance results reinforce the earlier findings by demonstrating that machine learning models are able to leverage a broader and more complex set of predictors, thereby improving predictive performance relative to traditional approaches.

Random Forest Variable Importance



The variable importance plot highlights key predictors driving the Random Forest model. Results confirm the importance of macroeconomic and financial indicators such as the yield curve and labor market variables.

VI. Discussion of Findings and Conclusion

The results of this study should be interpreted along two dimensions: statistical performance and economic usefulness. From a statistical perspective, the evaluation metrics—including AUC, recall, precision, and F1 score—provide evidence on whether machine learning models improve the classification of recession-risk periods relative to the logistic regression benchmark. From an economic perspective, the more important question is whether these models generate meaningful early-warning signals ahead of economic downturns.

Consistent with expectations, the logistic regression model serves as a transparent and interpretable benchmark that aligns with traditional recession forecasting literature. Its results reflect well-established relationships, particularly the predictive role of the yield curve spread and labor market conditions. However, its linear structure limits its ability to capture more complex dynamics across macroeconomic and financial variables.

In contrast, machine learning models—particularly Random Forest—demonstrate the ability to capture nonlinear relationships and interactions across predictors. For example, the impact of an inverted yield curve may vary depending on broader macro-financial conditions, such as elevated inflation, tightening monetary policy, or increased market volatility. This flexibility allows machine learning models to better adapt to changing economic environments.

The inclusion of financial market variables, such as equity returns, volatility (VIX), and oil price movements, enhances the model’s responsiveness to forward-looking information. Financial markets often incorporate expectations about future economic conditions more rapidly than traditional macroeconomic indicators. As a result, the model is able to indirectly capture the effects of geopolitical

events and inflationary pressures through observable market-based signals, rather than relying on subjective event-based variables. This approach strengthens the reproducibility and empirical consistency of the analysis.

A key challenge in recession prediction is the rarity of recession events, which creates class imbalance in the dataset. This makes accuracy an unreliable performance metric, as a model could achieve high accuracy by predominantly predicting non-recession periods. To address this issue, this study emphasizes metrics such as AUC, recall, precision, and calibration, which provide a more informative assessment of model performance in identifying recession signals.

Another important limitation relates to data availability and revisions. The dataset is constructed using the most recent observations from the Federal Reserve Economic Data (FRED) database. However, real-time forecasters would only have access to data available at each historical point in time. As a result, the model's performance may be somewhat optimistic relative to a true real-time forecasting setting.

Overall, the findings suggest that machine learning models can improve the detection of recession risk, particularly in capturing complex relationships across macroeconomic and financial variables. However, these improvements should be interpreted cautiously. Recession forecasting remains inherently difficult, especially during periods of structural change or unexpected shocks, such as the COVID-19 pandemic.

The most practical implication of this study is that machine learning should be viewed as a complementary tool rather than a replacement for traditional economic analysis. A robust recession-warning framework should integrate interpretable macroeconomic indicators, market-based signals, rigorous model validation, and informed economic judgment. This combined approach provides a more reliable foundation for understanding and anticipating economic downturns.

VII. References

Data Sources

- Federal Reserve Economic Data FRED.
- NBER-based recession indicators USREC.
- 10-Year Treasury Constant Maturity Minus 3-Month Treasury T10Y3M.
- Industrial Production Index INDPRO.
- Effective Federal Funds Rate FEDFUNDS.
- S&P 500 Index SP500.
- CBOE Volatility Index VIXCLS.
- Crude Oil Prices: West Texas Intermediate DCOILWTICO.
- Consumer Price Index for All Urban Consumers CPIAUCSL.
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